05

Lecture 05

4G LTE Architecture

معماری LTE نسل چهارم

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

DIFFICULTY
Intermediate

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

PREREQUISITES
Lectures 1–4; OFDM basics; IP networking

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 4
6 Mathematical models . . . 6
7 Network architecture and diagrams . . . 8
8 Step-by-step procedures . . . 9
9 Worked examples . . . 9
10 Technology comparison . . . 10
11 Key takeaways . . . 10
12 Exercises . . . 11
13 Further reading . . . 13

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-05-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the limitations of 3G that motivated LTE and list the main LTE requirements.
- **LO2** **State** the LTE design principles – flat, all-IP, shared channels, no soft handover – and the engineering reason for each.
- **LO3** **Draw** the Evolved Packet System (E-UTRAN + EPC) and **name** every reference point (S1-MME, S1-U, X2, S5/S8, S6a, S11, SGi, Gx, Rx).
- **LO4** **Derive** the LTE downlink peak rate from the resource grid and **build** a user-plane latency budget.
- **LO5** **Describe** how voice is carried in an all-IP system (IMS, VoLTE, CS fallback, SRVCC).
- **LO6** **Explain** control/user plane separation (CUPS) and its role as a bridge to 5G.

2 Prerequisite knowledge

You should know the UMTS architecture and its CS/PS split (Lecture 3), the GTP tunnel of GPRS (Lecture 1) and the basic idea of OFDM: a wideband channel divided into many narrow, orthogonal subcarriers, each protected from inter-symbol interference by a cyclic prefix.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
LTE	تکامل بلندمدت	3GPP radio access of 4G (E-UTRA), from Release 8.
EPS	سامانهٔ بستهٔ تکامل‌یافته	Complete 4G system: E-UTRAN + EPC.
eNB (eNodeB)	ایستگاه پایهٔ LTE	Base station incl. radio resource management; no RNC.
EPC	هستهٔ بستهٔ تکامل‌یافته	All-IP core: MME, S-GW, P-GW, HSS, PCRF.
MME	موجودیت مدیریت تحرک	Control-plane node: NAS, mobility, security, bearer control.
S-GW / P-GW	دروازهٔ خدمت‌دهنده / دروازهٔ PDN	User-plane anchors: mobility anchor / IP anchor.
EPS bearer	حامل EPS	End-to-end QoS pipe between UE and P-GW.
OFDMA / SC-FDMA	دسترسی چندگانهٔ متعامد / تک‌حامل	LTE downlink / uplink multiple access.
VoLTE	صدا روی LTE	IMS-based voice over the EPS bearer of QCI 1.
CUPS	جدایی صفحهٔ کنترل و کاربر	Rel-14 split of S/P-GW into C and U parts (PFCP).

3 Introduction and motivation

By 2004 HSPA was delivering mobile broadband, but its foundations were showing their age. WCDMA's 5 MHz carrier could not grow to the 10–20 MHz needed for higher rates without complex equalisers; the hierarchy UE–NodeB–RNC–SGSN–GGSN added latency and cost; two parallel cores (CS and PS) doubled operations. Meanwhile competing technologies (WiMAX) were promising OFDM-based, all-IP broadband.

3GPP's Long Term Evolution study set ambitious targets [8]: peak rates of 100 Mbit/s downlink and 50 Mbit/s uplink in 20 MHz; user-plane latency below 5 ms; idle-to-active transition below 100 ms; 2–4 times the spectral efficiency of Release-6 HSPA; scalable bandwidth from 1.4 to 20 MHz; and support for speeds up to 350 km/h. The *System Architecture Evolution* (SAE) defined a matching core, the EPC. Together they form the Evolved Packet System (EPS), first specified in Release 8 (2008) [3], [4].

4 Fundamental concepts

4.1 Design principles and their reasons

- **Flat RAN.** The RNC is removed; its functions move to the eNB (radio resource control, scheduling, ciphering, handover decision) or to the core. Fewer nodes means lower latency and cost. The price is that no node can combine signals from several cells – hence no soft handover (Lecture 4).
- **All-packet core.** There is no circuit-switched domain. Voice must be carried as IP packets (VoLTE over IMS), or temporarily handed to 2G/3G (CS fallback).
- **Shared channels only.** All user data uses shared channels scheduled every 1 ms by the eNB; there are no dedicated traffic channels.
- **OFDMA downlink, SC-FDMA uplink.** OFDM scales gracefully with bandwidth and makes frequency-selective scheduling and MIMO simple. In the uplink, single-carrier FDMA keeps a low peak-to-average power ratio for the handset's amplifier (Lecture 7).
- **Separation of control and user planes** in the core: the MME never touches user packets; the gateways never terminate NAS signalling.
- **Self-organising networks (SON).** Automatic neighbour relations, mobility robustness and load balancing reduce manual planning in a network of many small, flat nodes.

4.2 The EPS bearer

Definition 1 • EPS bearer

An EPS bearer is a logical pipe between the UE and the P-GW with a single QoS treatment. It is the concatenation of a *radio bearer* (UE–eNB), an *S1 bearer* (eNB–S-GW, a GTP-U tunnel) and an *S5/S8 bearer* (S-GW–P-GW, another GTP-U tunnel). Every UE gets a *default bearer* at attach; *dedicated bearers* are added for traffic needing different QoS, such as VoLTE (QCI 1).

Unlike GPRS, where connectivity required a separate PDP context activation after attach, LTE establishes the default bearer *during* attach: an LTE device is always on with an IP address. Bearers and QoS classes are developed in Lectures 9 and 22.

5 Detailed technical explanation

5.1 Network elements

UE. Mobile equipment plus UICC (with the USIM application). LTE UEs are classified by *categories* specifying peak rates, MIMO layers and buffer sizes.

eNB. The only node of the E-UTRAN [3]. It runs the full radio stack (PHY, MAC, RLC, PDCP, RRC), schedules uplink and downlink every millisecond, performs admission control, handover decisions, header compression and ciphering of user data, routes user data to the S-GW and selects the MME for a UE at attach. eNBs connect to each other over X2 and to the core over S1.

MME. The control-plane brain of the EPC: terminates NAS signalling and its security, authenticates the UE with vectors from the HSS, manages tracking areas and paging, selects S-GW and P-GW, and sets up, modifies and releases bearers through the S-GW. It is the subject of Lecture 12.

S-GW. Terminates S1-U and is the *local mobility anchor*: during inter-eNB handover only the S1-U tunnel changes. It buffers downlink data for idle UEs and triggers paging through the MME, and it anchors mobility to 2G/3G.

P-GW. The *IP anchor*: allocates the UE's IP address, is the point of attachment to a packet data network (identified by an APN), enforces policy and charging (PCEF), performs deep packet inspection and lawful interception. A UE may have connections to several P-GWs for different APNs (e.g. `internet` and `ims`). Lecture 13 studies both gateways.

HSS. The subscriber database, evolved from the HLR: subscription profile, AuC functions, current MME. Lecture 11.

PCRF. The policy and charging rules function [7]: decides QoS and charging rules per service data flow and pushes them to the P-GW over Gx; application functions such as the IMS P-CSCF request resources over Rx.

5.2 Reference points

Table 1. Main EPS reference points [4].

Reference point	Between	Protocol	Purpose
LTE-Uu	UE – eNB	RRC, PDCP/RLC/MAC/PHY	Radio interface
S1-MME	eNB – MME	S1AP over SCTP	UE context, bearers, paging, handover
S1-U	eNB – S-GW	GTP-U over UDP/IP	User data per bearer
X2	eNB – eNB	X2AP / GTP-U	Handover, data forwarding, interference coordination
S11	MME – S-GW	GTPv2-C	Session and bearer management
S5/S8	S-GW – P-GW	GTPv2-C + GTP-U	S5 within a network, S8 between home and visited
S6a	MME – HSS	Diameter	Authentication vectors, location update, subscription
SGi	P-GW – PDN	IP	To the Internet, IMS or corporate networks
Gx	PCRF – P-GW	Diameter	Policy and charging rules
Rx	AF – PCRF	Diameter	Application requests for QoS (e.g. VoLTE)
Gy/Gz	P-GW – OCS/OFCS	Diameter / file	Online / offline charging

Two patterns stand out. First, every control interface of the core is either GTPv2-C (between nodes that manage tunnels) or Diameter (towards databases and policy) – SS7 and MAP have disappeared from the EPC. Second, S8 is the *roaming* variant of S5: with *home routing* the visited S-GW tunnels traffic back to the home P-GW, so the home operator keeps control of IP addressing and charging.

5.3 Voice in an all-IP system

Without a CS domain, LTE offers three voice solutions:

- **VoLTE** – voice as an IMS service [6]. The UE registers with IMS over a default bearer to the `ims` APN (QCI 5 for SIP signalling). At call set-up the P-CSCF asks the PCRF over Rx for resources; the PCRF instructs the P-GW over Gx to create a dedicated GBR bearer with QCI 1 for the speech packets (Lecture 10).
- **CS fallback (CSFB)** – the UE is registered in the 2G/3G CS domain via the MME (SGs interface); for a call it is redirected to 2G/3G [2]. Simple, but call set-up is slower and data speed drops during the call.
- **SRVCC** – single radio voice call continuity: an ongoing VoLTE call is handed over to the CS domain when the UE leaves LTE coverage [9].

5.4 Control and user plane separation (CUPS)

In Release 8 the S-GW and P-GW each contained both control logic (GTP-C, policy, IP allocation) and packet forwarding. Release 14 split them [1]: SGW-C and PGW-C keep the control logic and program SGW-U and PGW-U through the *Sx* interface with the Packet Forwarding Control Protocol (PFCP) [5]. The user planes can then be scaled independently and placed close to the radio for low latency (edge computing), while control remains centralised. CUPS is the direct ancestor of the 5G SMF/UPF split (Lectures 13 and 19).

6 Mathematical models

6.1 Downlink peak rate from the resource grid

Problem. Where does the "300 Mbit/s" of a Category-5 LTE device come from?

System model. Bandwidth 20 MHz, normal cyclic prefix, FDD. The resource grid (Lecture 7) has $N_{\mathrm{RB}} = 100$ resource blocks of 12 subcarriers at $\Delta f = 15\,\mathrm{kHz}$, and 14 OFDM symbols per 1 ms subframe, i.e. $12 \times 14 = 168$ resource elements (REs) per RB per subframe. Assumptions: one OFDM symbol for the control region, four cell-specific reference-signal ports, 64QAM ($Q_m = 6$), $\nu = 4$ MIMO layers, and we neglect the synchronisation and broadcast channels, which occupy a few subframes only.

Derivation. Four CRS ports use 24 REs per RB pair, four of which lie in the control symbol; the data REs per RB per subframe are therefore

$$N_{\mathrm{RE}}^{\mathrm{data}} = 168 - \underbrace{12}_{\text{control}} - \underbrace{20}_{\text{CRS}} = 136. \tag{1}$$

The peak rate is

$$R_{\mathrm{peak}} = \nu\, Q_m\, r_{\mathrm{c}}\, \frac{N_{\mathrm{RB}}\, N_{\mathrm{RE}}^{\mathrm{data}}}{T_{\mathrm{sf}}} = 4 \times 6 \times r_{\mathrm{c}} \times \frac{100 \times 136}{1\,\mathrm{ms}} = 326.4\, r_{\mathrm{c}}\ \mathrm{Mbit/s}, \tag{2}$$

where r_{c} is the channel-code rate and T_{sf} the subframe duration. With the highest standardised transport block ($r_{\mathrm{c}} \approx 0.92$) this gives ≈ 300 Mbit/s, consistent with the 299.6 Mbit/s of UE Category 5.

Interpretation. The reference signals and control region cost about 19% of the REs; the cyclic prefix (6.7% of time, Exercise 3) is already built into the 14-symbol subframe. Peak rates need four receive

antennas, excellent SINR and an empty cell – they are an upper bound of the air interface, not a user experience.

6.2 User-plane latency budget

Problem. Can LTE meet the 5 ms one-way user-plane latency target?

Model. One-way latency from the arrival of a packet at the eNB's PDCP to its delivery at the UE's IP layer, unloaded cell, decomposed as

$$D_{\mathrm{UP}} = \underbrace{\tfrac{1}{2}T_{\mathrm{TTI}}}_{\text{alignment}} + T_{\mathrm{TTI}} + D_{\mathrm{eNB}} + D_{\mathrm{UE}} + p_{\mathrm{re}}\, T_{\mathrm{HARQ}} + D_{\mathrm{tr}}, \tag{3}$$

where the alignment term is the mean wait for the next TTI boundary, D_{eNB} and D_{UE} are processing delays, p_{re} is the probability of one HARQ retransmission, $T_{\mathrm{HARQ}} = 8$ ms the FDD HARQ round trip and D_{tr} the residual transport delay.

Numerical example. With $T_{\mathrm{TTI}} = 1$ ms, $D_{\mathrm{eNB}} = D_{\mathrm{UE}} = 1$ ms, $p_{\mathrm{re}} = 0.1$ and $D_{\mathrm{tr}} = 0.5$ ms (illustrative values, Figure 1): $D_{\mathrm{UP}} = 0.5 + 1 + 1 + 1 + 0.8 + 0.5 = 4.8$ ms.

Interpretation and limits. The 1 ms TTI and the 8 ms HARQ loop dominate: these are the levers that 5G pulls (shorter slots, mini-slots, faster HARQ, Lecture 14). The model ignores queueing (zero load), scheduling-request delay in the uplink (a UE without a grant must first ask for one, adding several milliseconds) and core and Internet delay. Measured end-to-end ping times in LTE networks are therefore typically tens of milliseconds even though the radio target is 5 ms.

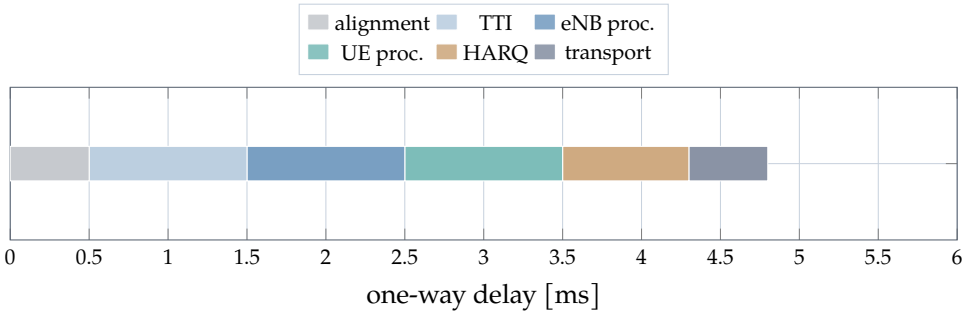


Figure 1. Illustrative one-way downlink user-plane latency budget of Eq. (3) (4.8 ms).

6.3 Encapsulation overhead

Every user packet crossing S1-U or S5 is carried in GTP-U/UDP/IPv4, adding $h = 8 + 8 + 20 = 36$ bytes. For a payload of L bytes the transport efficiency is

$$\eta_{\mathrm{tr}} = \frac{L}{L + h}. \tag{4}$$

A 1400-byte web packet has $\eta_{\mathrm{tr}} = 97.5\%$; a 60-byte VoLTE packet (speech frame plus RTP/UDP/IP headers; header compression acts only on the radio link) only 62.5%. Backhaul dimensioning for voice must account for this (Exercise 4); IPv6 transport adds a further 20 bytes.

7 Network architecture and diagrams

Figure 2 shows the EPS. Compared with UMTS (Lecture 3, Figure 3) the RAN has one node type, the core has no CS domain, and the user plane crosses only two core nodes (S-GW and P-GW), both of which can be co-located. Control (dashed) and user (solid teal) planes use different interfaces everywhere except the radio. Figure 3 shows the user-plane stack: radio protocols end in the eNB, and from there GTP-U tunnels carry the user's IP packets.

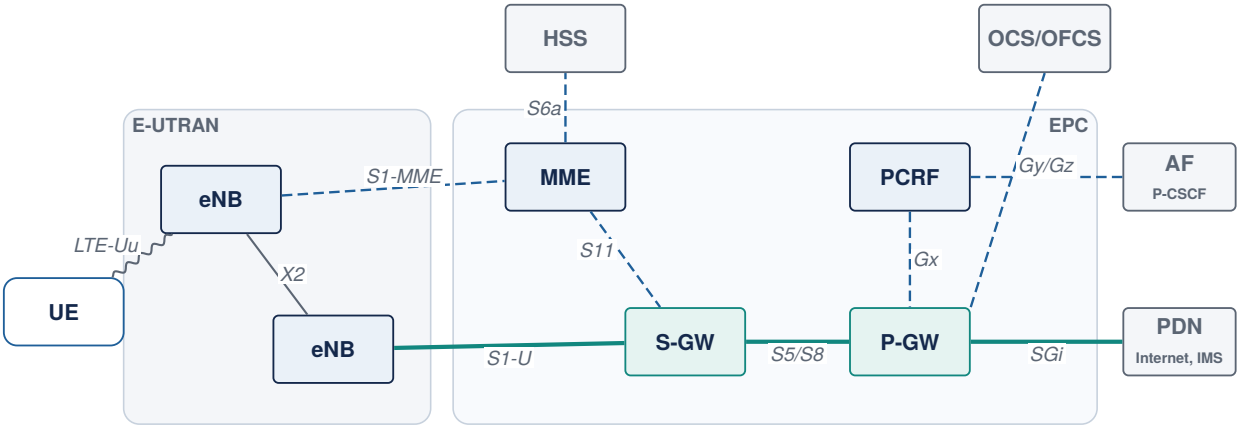


Figure 2. The Evolved Packet System. Dashed blue: control plane; solid teal: user plane.

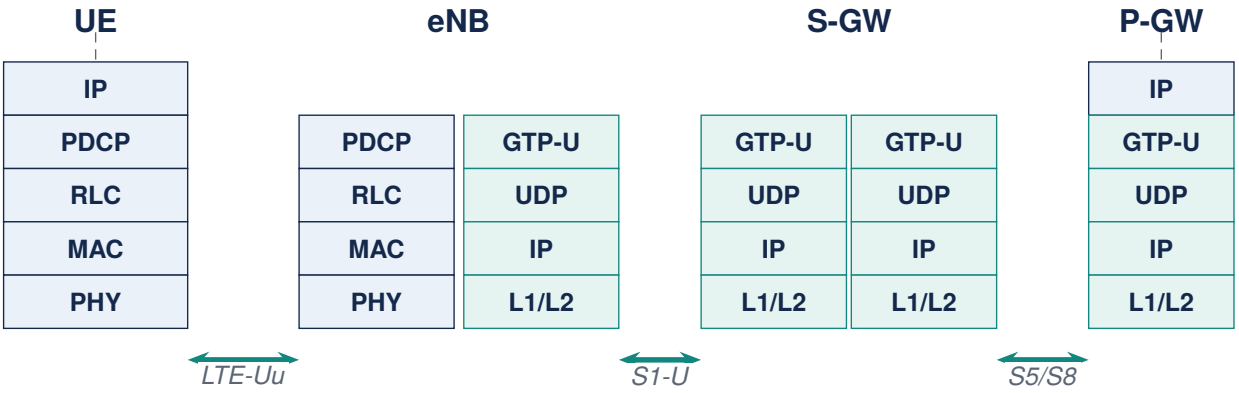


Figure 3. LTE user-plane protocol stack. The UE's IP packet is carried by PDCP/RLC/MAC over the radio and by GTP-U tunnels on S1-U and S5/S8; the P-GW is the IP anchor.

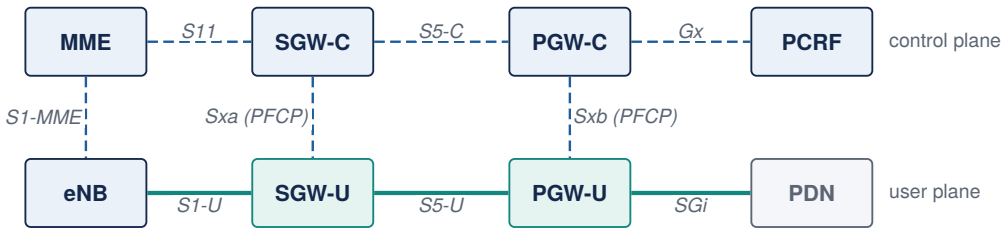


Figure 4. Control and user plane separation of the EPC gateways (Release 14).

8 Step-by-step procedures

Procedure 1 • LTE initial attach (overview)

1. The UE performs random access and sets up an RRC connection with the eNB (Lectures 6 and 8).
2. It sends a NAS Attach Request with a PDN Connectivity Request; the eNB selects an MME and forwards it in S1AP Initial UE Message.
3. The MME authenticates the UE with EPS-AKA (vectors from the HSS over S6a), activates NAS security, and registers itself in the HSS (Update Location), receiving the subscription.
4. The MME selects an S-GW and a P-GW and sends GTPv2-C Create Session Request; the P-GW allocates an IP address, obtains policy from the PCRF and creates the default bearer.
5. The MME sends S1AP Initial Context Setup Request with Attach Accept; the eNB activates AS security and configures the radio bearer with RRC Connection Reconfiguration.
6. The UE answers Attach Complete; the MME sends Modify Bearer Request so that the S-GW learns the eNB's tunnel endpoint. Uplink and downlink data can flow.

The detailed message flow, with every information element that matters, is built up in Lectures 9, 11 and 12. Note already that one procedure delivers registration, security *and* an IP address – three separate GPRS procedures merged into one.

9 Worked examples

Worked example 1 • Peak rate of a 10 MHz cell

Problem. A 10 MHz LTE carrier (50 RBs) uses two CRS ports, two control symbols and 2×2 MIMO with 64QAM. Estimate the peak rate for $r_c = 0.9$.
Solution. Two CRS ports use 16 REs per RB pair, of which 4 lie in the first (control) symbol; with two control symbols, $N_{\mathrm{RE}}^{\mathrm{data}} = 168 - 24 - 12 = 132$. $R = 2 \times 6 \times 0.9 \times 50 \times 132/10^{-3} = 71.3\,\mathrm{Mbit/s}$.
Interpretation. Half the bandwidth and half the layers give roughly a quarter of the Category-5 figure; the control region and RS overhead are similar ($\approx 21\%$).

Worked example 2 • Hops in the user plane

Problem. Count the network nodes that process a user packet between the radio and the Internet in UMTS R99 and in LTE, and estimate the saving if each node adds 0.5 ms of processing and queuing.
Solution. UMTS: NodeB, RNC, SGSN, GGSN (4 nodes). LTE: eNB, S-GW, P-GW (3 nodes; 2 if S-GW and P-GW are combined). Saving: 0.5–1 ms one way, 1–2 ms round trip.
Interpretation. The node count matters less than the RNC's long TTIs and slow scheduling, which the eNB replaces by 1 ms scheduling – the main reason LTE's latency is a fraction of R99's.

Worked example 3 • VoLTE capacity of a cell

Problem. A VoLTE call needs on average 2 RBs every 20 ms in each direction (one speech frame per 20 ms, robust MCS). How many calls can a 10 MHz cell carry if 70% of RBs may be used for voice?

Solution. Per call: 2 RBs per 20 subframes = 0.1 RB per subframe. Available: $0.7 \times 50 = 35$ RBs per subframe. Calls $= 35/0.1 = 350$ per direction (in practice the PDCCH control channel limits this earlier, which is why *semi-persistent scheduling* is used for VoLTE).

Interpretation. Compare with about 45 WCDMA voice users per 5 MHz (Lecture 3): packet voice with header compression and scheduling is far more spectrum-efficient than circuit voice.

10 Technology comparison

Table 2. 3G (UMTS/HSPA) versus 4G (LTE/EPC).

	UMTS / HSPA	LTE / EPC
Radio access	WCDMA, 5 MHz	OFDMA DL / SC-FDMA UL, 1.4–20 MHz
RAN nodes	NodeB + RNC	eNB only (flat)
Handover	Soft + hard	Hard (with X2 forwarding)
Core domains	CS + PS	PS only
Voice	CS	VoLTE (IMS); CSFB/SRVCC for interworking
Core signalling	MAP (SS7), GTPv1-C, RANAP	Diameter, GTPv2-C, S1AP
TTI	2 ms (HSPA), 10–80 ms (R99)	1 ms
Always-on IP	No (PDP context on demand)	Yes (default bearer at attach)
Peak DL (typical device)	14–42 Mbit/s	150–300 Mbit/s (Rel-8)

Trade-offs. The flat, all-IP design simplified the network but moved complexity elsewhere: QoS for voice must now be engineered with bearers and policy (PCRF), the eNB carries more functions and state, mobility without soft handover relies on fast hard handovers, and the transition period needed CSFB and SRVCC. The control/user plane separation already present in the EPC (MME versus gateways) and completed by CUPS foreshadows the service-based 5G core.

11 Key takeaways

Key takeaways

- LTE requirements: 100/50 Mbit/s peak in 20 MHz, <5 ms user-plane latency, scalable bandwidth, packet-only.
- EPS = E-UTRAN (eNB only) + EPC (MME, S-GW, P-GW, HSS, PCRF).

- Control interfaces use S1AP, GTPv2-C and Diameter; user data uses GTP-U on S1-U and S5/S8.
- S-GW = mobility anchor; P-GW = IP anchor and policy enforcement; MME = signalling only.
- A default EPS bearer is created at attach: always-on IP.
- Peak rate $\approx \nu Q_m r_c N_{\mathrm{RB}} N_{\mathrm{RE}}^{\mathrm{data}} / T_{\mathrm{sf}}$; the 1 ms TTI and 8 ms HARQ loop dominate the radio latency.
- Voice: VoLTE over IMS with a QCI 1 bearer; CSFB and SRVCC for interworking with 2G/3G.
- CUPS splits the gateways into C and U parts controlled with PFCP – the bridge to 5G.

12 Exercises

Exercise 1 • Level 1 – conceptual • Why two gateways?

Explain why the EPC has both an S-GW and a P-GW. Give one deployment where they are combined and one where they must be separate.

Exercise 2 • Level 1 – conceptual • Where did the RNC go?

For each of the following RNC functions state where it lives in LTE: RRC termination, ciphering of user data, handover decision, outer-loop power control, mobility anchoring towards the core.

Exercise 3 • Level 2 – numerical • Cyclic-prefix overhead

In LTE with normal CP, a 0.5 ms slot contains 7 OFDM symbols of $2048\,T_{\mathrm{s}}$ with $T_{\mathrm{s}} = 1/(15000 \times 2048)$ s; the first CP lasts $160\,T_{\mathrm{s}}$ and the others $144\,T_{\mathrm{s}}$. Compute the CP durations in microseconds and the fraction of time spent on CP. What is the maximum delay spread tolerated without inter-symbol interference?

Exercise 4 • Level 2 – numerical • VoLTE backhaul

A VoLTE packet carries a 33-byte speech frame every 20 ms with 12 bytes RTP, 8 bytes UDP and 20 bytes IPv4. On S1-U it is encapsulated in GTP-U/UDP/IPv4. Compute the bit rate per call on the backhaul and the transport efficiency Eq. (4). How many calls fit in a 100 Mbit/s link loaded at 70%?

Exercise 5 • Level 3 – analytical • Uplink latency

Extend Eq. (3) to the uplink of a UE that has no grant: it must wait for its next scheduling-request (SR) opportunity (period T_{SR}), send the SR, receive a grant after k_1 ms and transmit k_2 ms after the grant. Derive the mean latency and evaluate it for $T_{\mathrm{SR}} = 10$ ms, $k_1 = 3$ ms, $k_2 = 4$ ms, other values as in Section 6.2.

Exercise 6 • Level 3 – analytical • Home routing versus local breakout

A subscriber from Tehran roams in Istanbul and browses a server in Istanbul. Compare the user-plane path and the round-trip propagation delay for home-routed roaming (S8 to the home P-GW) and local breakout (visited P-GW). Assume 2000 km between the cities, fibre at 2×10^8 m/s and a route factor of 1.5. What does each option mean for charging and lawful interception?

Solutions

Solution to Exercise 1

They anchor different things. The S-GW anchors mobility between eNBs (and to 2G/3G) and buffers for idle UEs; there is one per UE at a time and it may change with location. The P-GW anchors the IP address and the connection to a PDN and enforces policy; it must not change during a session. Combined: a small national operator, one site, no roaming. Separate: roaming with home routing (visited S-GW, home P-GW over S8), or a UE with an Internet PDN on a local P-GW and an IMS PDN on a central P-GW.

Solution to Exercise 2

RRC termination: eNB. User-data ciphering: eNB (PDCP). Handover decision: source eNB. Outer-loop power control: not needed in the same form – LTE uses scheduled, orthogonal resources; uplink power control (open loop plus closed-loop corrections) is in the eNB. Mobility anchoring towards the core: S-GW (the RNC's Iu anchor role).

Solution to Exercise 3

$T_s = 32.55$ ns. First CP: $160\,T_s = 5.21\,\mu$s; others: $144\,T_s = 4.69\,\mu$s; useful symbol $2048\,T_s = 66.67\,\mu$s. CP fraction: $(160 + 6 \times 144)/15360 = 6.67\%$. Delay spreads up to about $4.7\,\mu$s (about 1.4 km of path difference) are absorbed; the extended CP ($512\,T_s = 16.7\,\mu$s) serves large cells and broadcast.

Solution to Exercise 4

IP packet: $33 + 12 + 8 + 20 = 73$ bytes; on S1-U: $73 + 36 = 109$ bytes every 20 ms $= 43.6$ kbit/s (plus Ethernet framing in practice). Efficiency relative to the speech payload: $33/109 = 30\%$; relative to the user IP packet Eq. (4): $73/109 = 67\%$. Calls: $0.7 \times 100 \times 10^6/43\,600 \approx 1600$ per direction.

Solution to Exercise 5

Mean wait for an SR opportunity $T_{SR}/2$, then SR transmission (1 ms), grant after k_1, data k_2 after the grant, then the downlink-like terms: $D_{UL} = T_{SR}/2 + T_{TTI} + k_1 + k_2 + T_{TTI} + D_{eNB} + p_{re}T_{HARQ} + D_{tr} = 5 + 1 + 3 + 4 + 1 + 1 + 0.8 + 0.5 = 16.3$ ms. The scheduling-request cycle dominates; pre-scheduling or short SR periods reduce it, and 5G adds configured grants that remove it.

Solution to Exercise 6

Home routing: Istanbul UE $\rightarrow$ visited S-GW $\rightarrow$ S8 to Tehran P-GW $\rightarrow$ back to the Istanbul server: two traversals of $1.5 \times 2000 = 3000$ km each way, i.e. 12 000 km per round trip, or 60 ms

of propagation. Local breakout: essentially local, $\ll 1$ ms. Home routing keeps charging, policy and lawful interception in the home network (simple for the home operator); local breakout requires the visited network to apply the home operator's policy (via S9 / roaming agreements) and to perform interception locally.

13 Further reading

The E-UTRAN overall description is [3] and the EPC procedures [4]; the LTE requirements are in [8]. Policy and charging control is specified in [7], CUPS in [1] with PFCP in [5], CSFB in [2] and SRVCC in [9]. Sesia et al. [11] and Olsson et al. [10] are the recommended textbooks for the radio and core sides respectively.

References

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